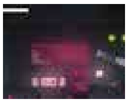

PewDiePie vs T-Series gets a game
Felix "PewDiePie" Kjellberg's subscription battle has been turned into a video game Zero Death. **https://dgit.in/apr19-95**

Angry Birds AR mobile game
Rovio is now bringing the Angry Birds AR: Isle of Pigs to iOS-powered devices with ARKit support. **https://dgit.in/apr19-96**

elements required for their production, such as cobalt. The prices of the raw materials for making the batteries are already soaring.

ASSORTED COMPONENTS

There are a bunch of assorted chips that go into a device, for various functions such as controlling the charge to the battery, USB charging and power management. There are also transceivers, modems, and connectivity modules for Bluetooth, WiFi and for cellular connectivity such as 4G. All these chips are typically made from a few semiconductor companies. These are Texas Instruments, Broadcom, Qualcomm, NXP, Skyworks, Maruta, Maxim, Qorvo, Micron and Cypress. The audio chips, at least in the high end phones, are mostly provided by Cirrus Logic, Qualcomm and HiSilicon. All these chips are for the most part, located on the motherboard, or the SoC. The other components such as the processor and the RAM chips plug into the motherboard. The motherboard of the device dictates how much real estate is available for all the components, and over the generations, these have become increasingly narrow strips, with the components packed in more and more densely. Most of the space within a device is occupied by the battery, and as battery life remains a pain point for smartphones (feature phones can last weeks), this is an area of rapid evolution. Sometimes the companies can push too far, and leave very little space for batteries. Lithium ion batteries expand and contract over the course of the charging cycles, and so need some breathing room within the device. If this breathing room is not enough, well then the devices can explode.

Apple uses just about the densest motherboards, which are actually folded up and is a 3D structure, with vents providing wiring access between the components on two layers. Apple calls them "logic boards". The iPhone X has two batteries to optimally use the space occupied by the roughly L shaped logic board. In fact, the density of components on the iPhone X logic board is more than that of the Apple Watch. Some other companies have other creative solutions, including secondary daughter boards for a discrete set of components. This is an innovative space saving measure. Vivo and Huawei are among the companies that use these daughter boards within their devices. There are heat pipes, RF shields, thermal pads for insulation to prevent your fingers from heating up even though it does so on the inside. The thermal insulation and heat sinks are particularly required for devices that support wireless charging, which tends to generate a lot of heat.